Area and Perimeter — Finding and Fixing Area/Perimeter Mix-Ups

Grades 2–3

Two questions sound almost the same but ask for different things.

- **Perimeter** is the distance all the way *around* the outside. You **add** the sides. The answer is a length: cm or m.
- **Area** is how many unit squares *cover* the inside. For a rectangle you **multiply** the long side by the short side. The answer is a number of squares: cm^2 or m^2.
- Before you compute, underline the words in the question that tell you which one it wants: *around, fence, ribbon, edging* means perimeter; *cover, fill, tiles, space inside* means area.

1. **Perimeter.** A name card is 8 cm long and 3 cm wide. Ali glues ribbon all the way around the outside edge of the card. How many cm of ribbon does she use?

Answer: ________________ cm

2. **Area.** The same name card is 8 cm long and 3 cm wide. Ali covers the whole card with one-centimetre squares. How many squares does it take?

Answer: ________________ cm^2

3. Find and fix. Milo needs the *perimeter* of a photo that is 6 cm long and 4 cm wide. He wrote $6 \times 4 = 24$ and said the perimeter is 24 cm.

Which measurement did Milo actually find? Cross out his work, then find the perimeter the right way.

Answer: ________________ cm

4. Find and fix. Sara needs the *area* of a rug that is 9 m long and 2 m wide. She wrote $9 + 2 + 9 + 2 = 22$ and said the area is 22 square m.

Which measurement did Sara actually find? Cross out her work, then find the area the right way.

Answer: ________________ m^2

5. Fence around a square. A square garden has four sides that are all the same length. Each side is 7 m long. A fence goes all the way around the garden. How many m of fence are needed?

Answer: ________________ m

6. Which one does the question want? Mr. Ruiz has a rectangular play mat that is 10 m long and 6 m wide. He wants soft edging along every outside edge of the mat.

First circle the word that names what the question wants: **perimeter** / **area**. Then find how many m of edging he needs.

Answer: ________________ m

7. Same fence, different space. Ana's garden is a square with sides of 5 m. Ben's garden is a rectangle 8 m long and 2 m wide. Both gardens need exactly the same length of fence around them — check that for yourself first.

How many more square m of growing space does Ana's garden have than Ben's?

Answer: ________________ m^2

8. Show what you know. On the grid below:

(a) Draw two *different* rectangles whose perimeters are the same but whose areas are different. Use whole grid squares.
(b) Under each rectangle write its perimeter and its area, with the correct kind of unit for each.
(c) Write one sentence a classmate could use to decide whether a question is asking for area or for perimeter.